

Single-GPU CG solve time versus matrix dimension

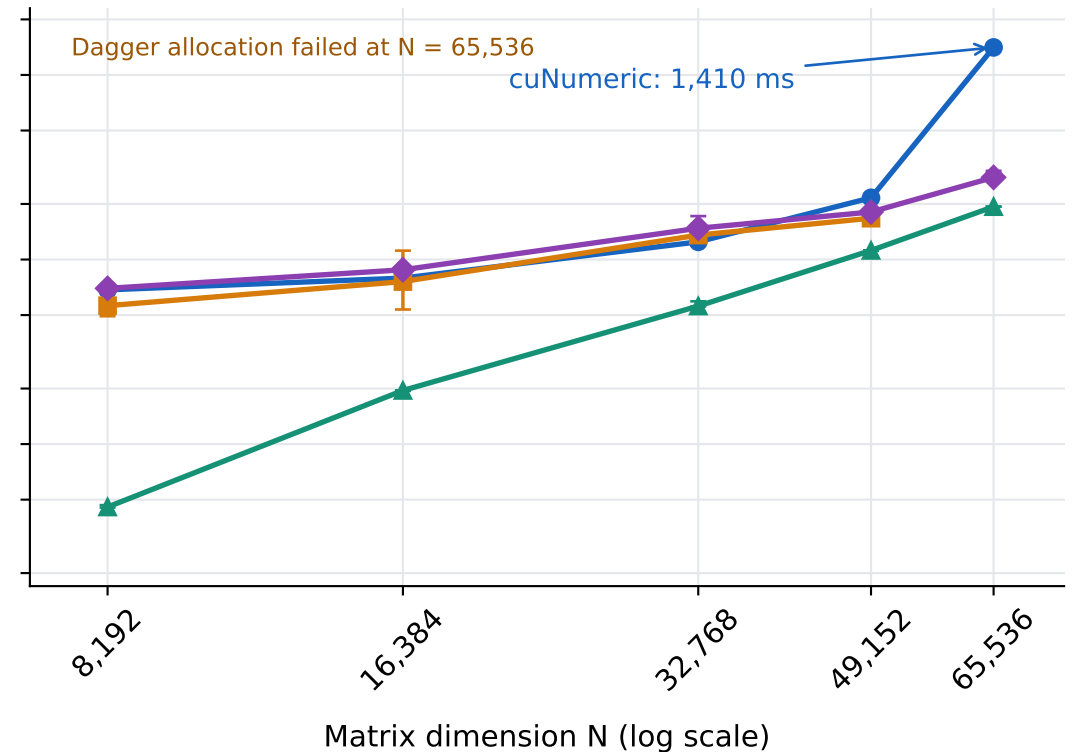
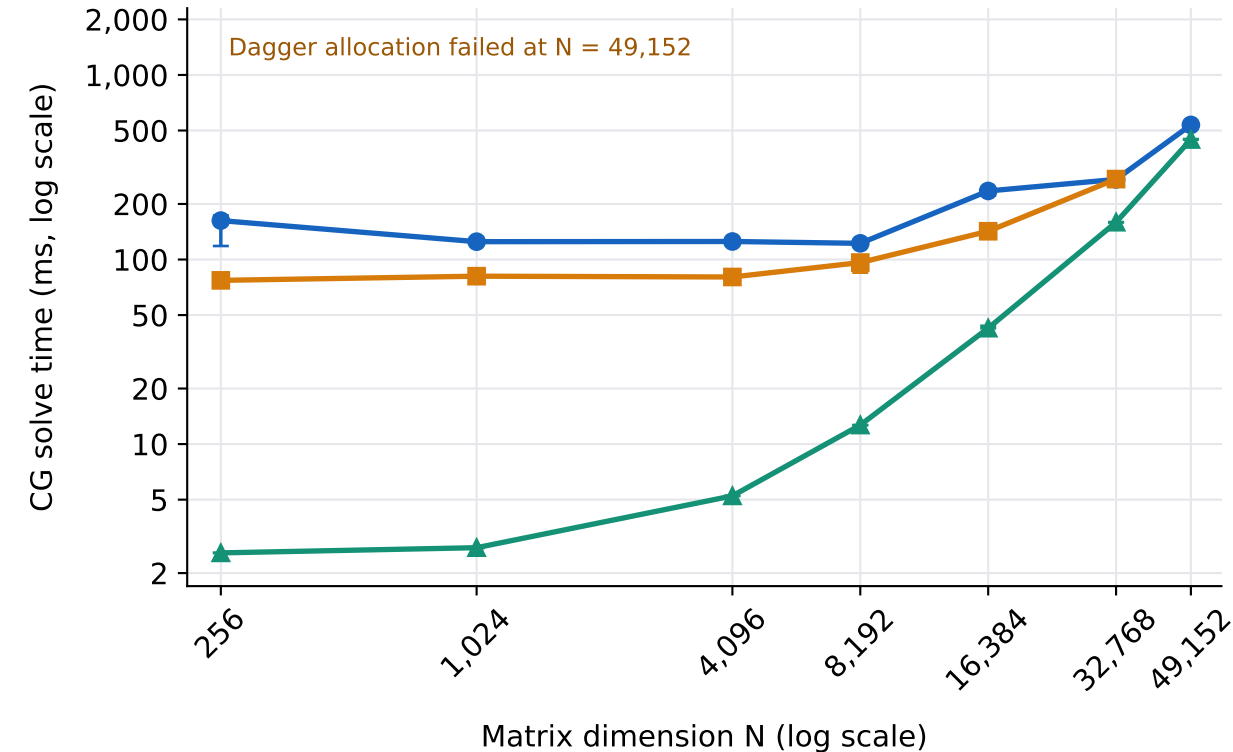
NVIDIA A30X · Dense $N \times N$ matrices · Median of 5 warmed solves

cuNumeric
Dagger (matvec workaround)

CUDA / CuArray
cuNumeric (transposed storage)

FP64 · $\text{rtol} = 10^{-8}$ · 18-19 iterations

FP32 · $\text{rtol} = 10^{-5}$ · 11-12 iterations



Whiskers show sample min-max (not confidence intervals). Setup and transfers excluded; solve completion synchronized.
Compare backends within each precision: tolerances and iteration counts differ. All successful solves passed residual checks.
GPU memory allowance increased for larger cases; full settings and raw samples are recorded alongside this figure.